

Client 2: Deployable Questionnaire

Administration-ready questionnaire guide with universe rules, timing, and row-level rationale

Prepared for client review | Updated: April 2026 | <https://github.com/namo507/Team-PSID>

Abstract

This operational report converts the scored PSID crisis-module output into a field-ready questionnaire with routing rules, timing estimates, and row-level rationale. It explains how the 28-row scored module becomes a 26-question deployable instrument, how respondent answers activate only one relevant crisis block, and how the final field package preserves analytic traceability while reducing burden.

Keywords: deployable questionnaire | crisis routing | respondent burden | PSID module | field administration

1 Purpose

This report is the PDF-ready version of the deployable questionnaire package. It translates the final scored question bank into a practical administration guide with live question tables, universe rules, timing estimates, and the ranking logic behind selection.

Deployable scope

This deployable version begins from the same 28-row scored module used in the technical review package. It excludes `SPC_AGE_YEARS` (0.35 minutes) and `SPC_DESCRIBE_SEX_GENDER` (0.817 minutes), which remain in the master review file for traceability but are removed from routine field administration. The result is a 26-question instrument with an estimated paper length of $29.98 - 0.35 - 0.817 = 28.82$ minutes.

2 Deployable Snapshot

Metric	Value	Calculation / rule	Why it is used and the assumption behind it
Deployable questions	26	$28 - 2 = 26$ after removing the selected demographic-only rows	Defines the paper instrument that can be handed to field staff.
Estimated deployable minutes	28.82	$29.98 - 0.35 - 0.817 = 28.813 \approx 28.82$	Upper-bound paper length if the full deployable document is read end to end.
All-respondent Generic questions	3	Count of rows asked before any routing logic is applied	Portable screening core; assumes every respondent should receive the same baseline crisis probe.
Pandemic or public-health questions	5	Count of rows in the pandemic block	Activated only when respondent answers or interview context indicate a pandemic/public-health event.
Shutdown or financial-disruption questions	4	Count of rows in the shutdown/financial-disruption block	Activated only when the crisis is economic or shutdown driven.
Disaster, displacement, or direct-exposure questions	14	Count of rows in the disaster/displacement block	Largest block because direct-exposure content carries the richest disaster signal in the source bank.

Table 1: Deployable instrument summary after removing demographic-only rows.

3 Figures and Visual Diagnostics

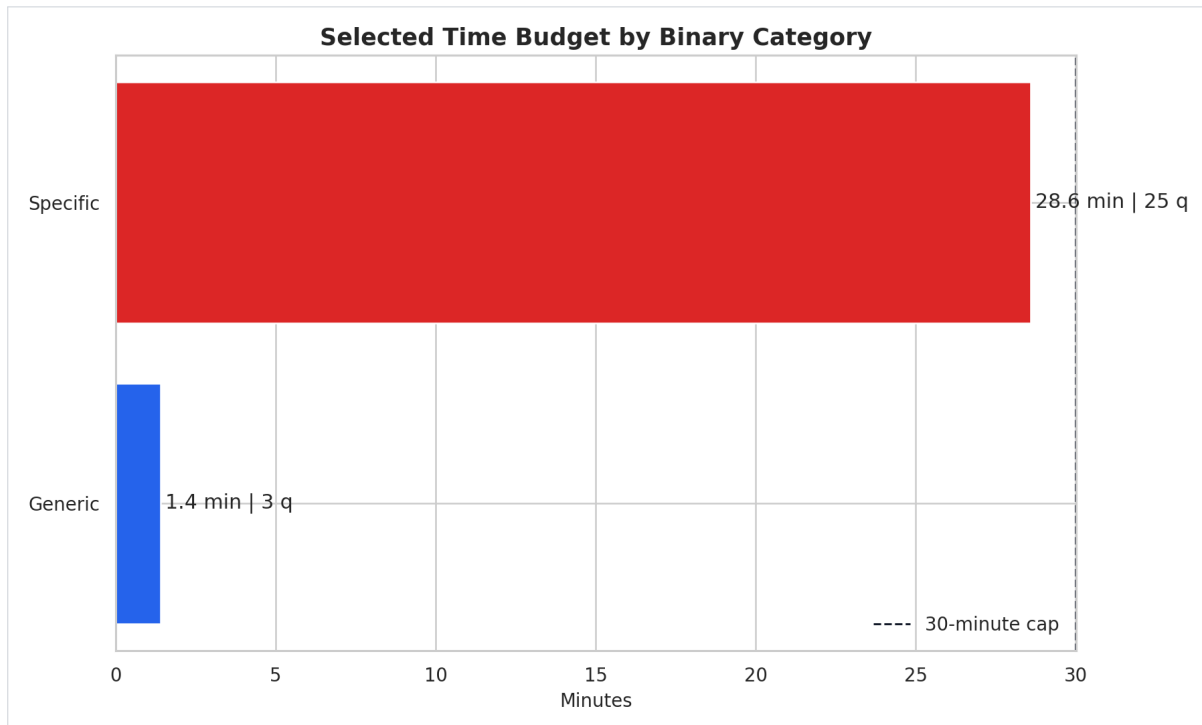


Figure 1: Time-budget allocation across the selected scored module.

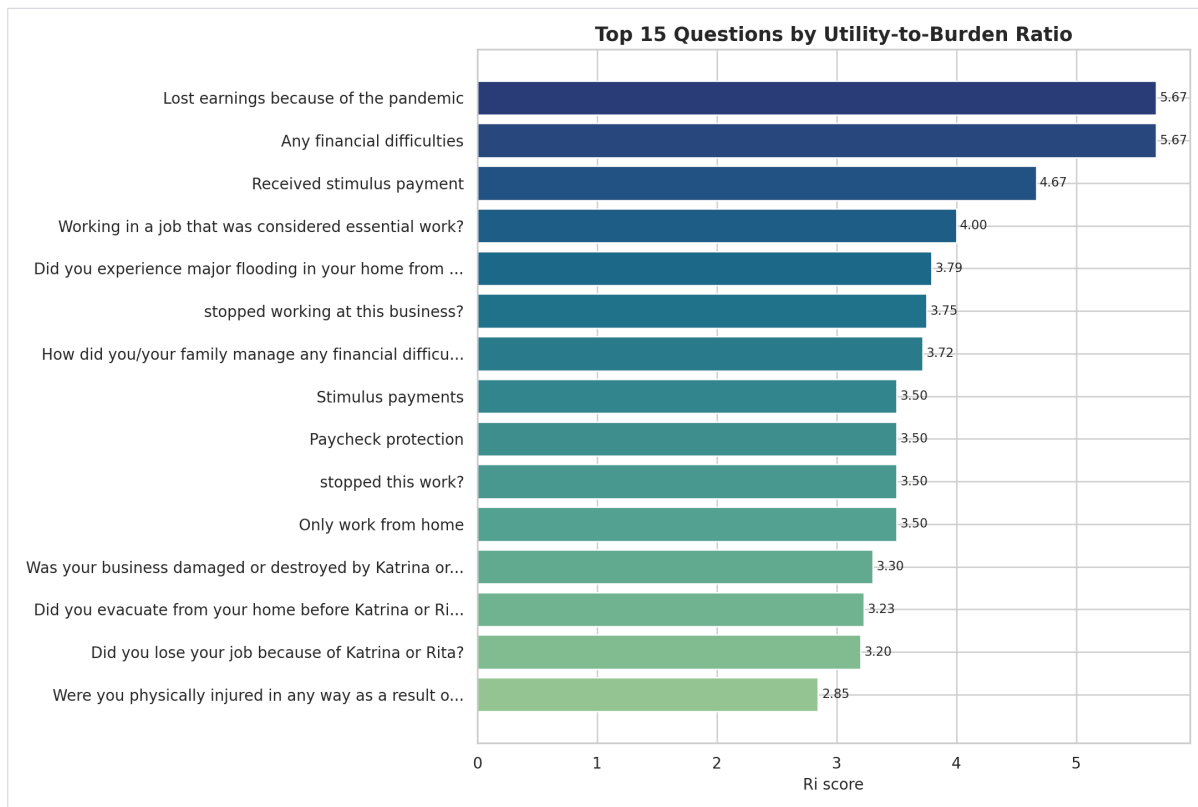


Figure 2: Top-ranked items retained in the refreshed package.

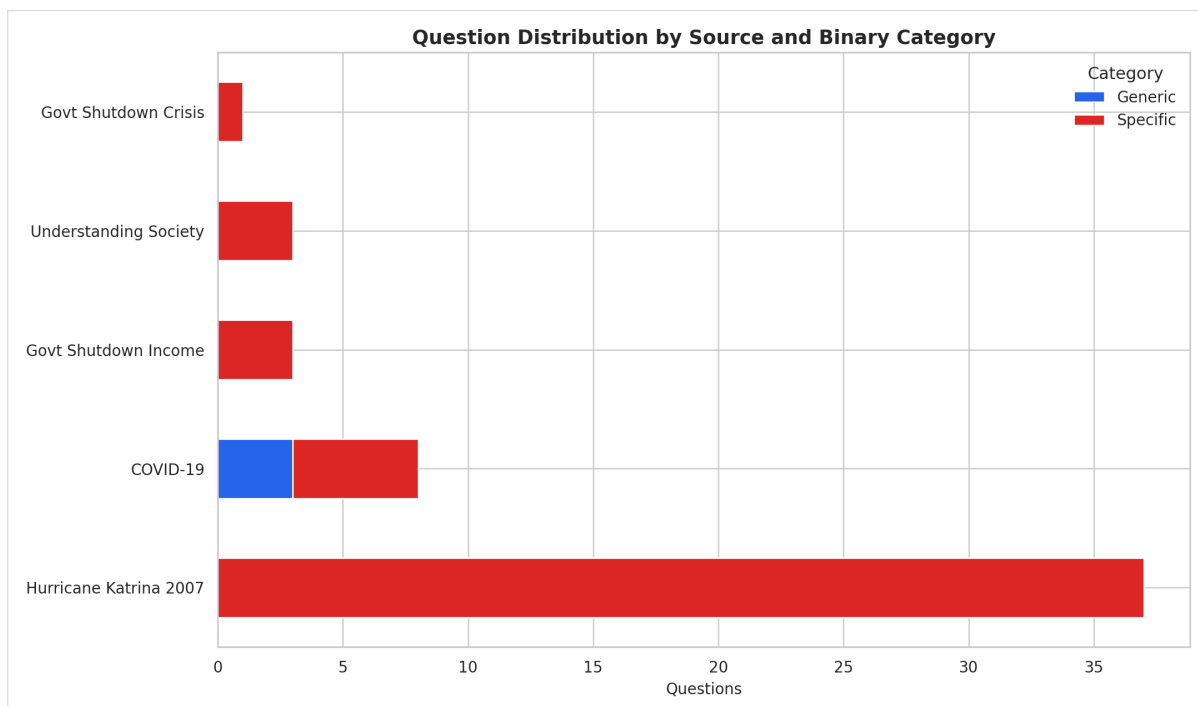


Figure 3: Binary Generic versus Specific split used to route administration.

4 Deployable Workflow Pipeline

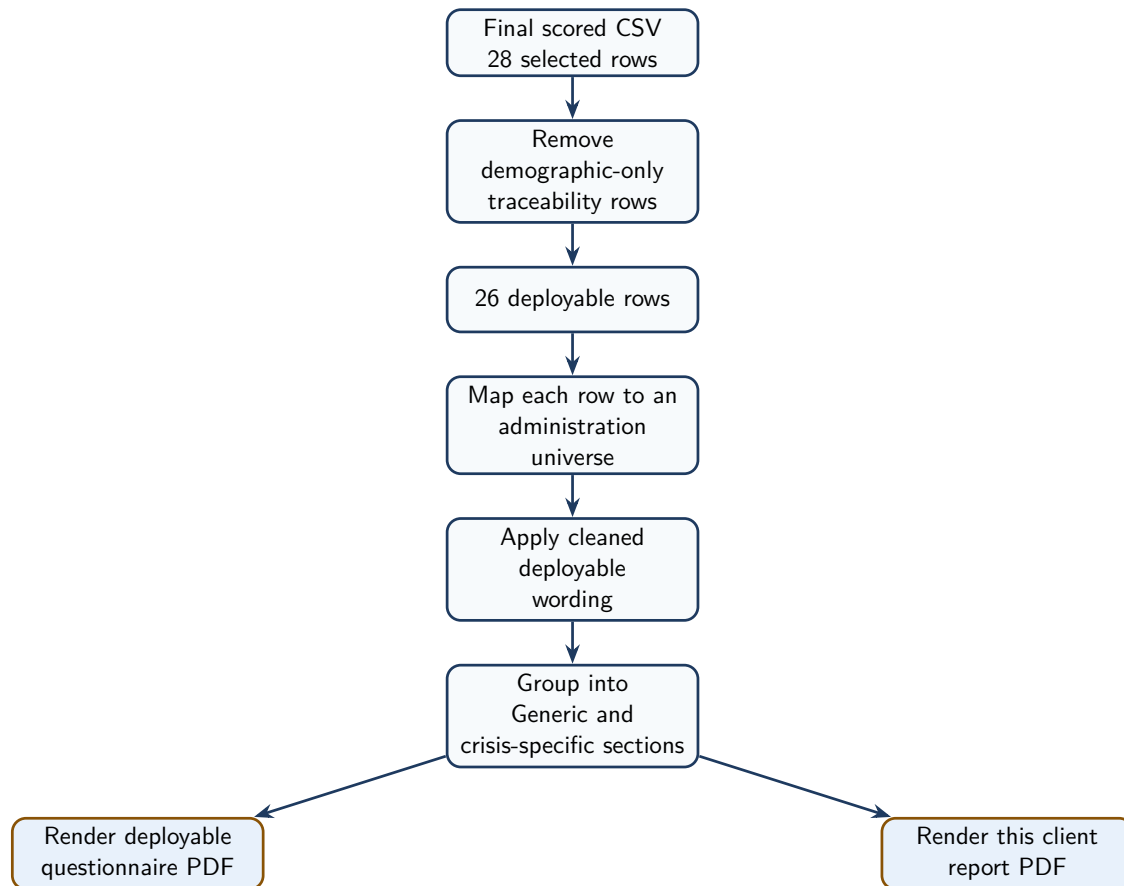


Figure 4: Operational path from the scored module to the deployable questionnaire package.

5 Why 28 Scored Rows Become 26 Asked Questions

Stage	Rows	Minutes	Calculation / rule	Operational meaning
Scored module	28	29.98	Count and summed minutes for rows with <code>selected_for_module = True</code>	Full analytic shortlist retained for auditability and review.
Selected demographic traceability rows	2	1.17	$\text{SPC_AGE_YEARS (0.35)} + \text{SPC_DESCRIBE_SEX_GENDER (0.817)}$	Context variables retained in the master file but not asked by default in the crisis instrument.
Deployable paper instrument	26	28.82	$28 - 2 = 26$ and $29.98 - 1.167 = 28.813 \approx 28.82$	Clean field-ready questionnaire containing only direct crisis items plus the universal Generic screen.

Table 2: The 28-to-26 change is a deployment-stage subtraction, not a change in the scoring model.

6 Optimization Change From Scored Review to Live Administration

Dimension	Legacy architecture view	Current deployable view	Operational effect
Category structure	Legacy architecture used 7 Generic Core rows, 1 Financial Crisis row, and 20 Pandemic / Disaster rows in the 28-row architecture	Current deployment asks 3 universal Generic rows first, then one routed crisis-specific block from a 23-row Specific bank	The universal screen is shorter and the field instrument is easier to explain.
Financial items	Financial questions were visible as their own toggle label	Financial questions now compete inside Specific under the same redundancy and portability logic as other crisis items	Duplicate earnings or aid variants no longer dominate the field package.
Deployment target	The scored shortlist and the field-facing package were closer together	The 28-row scored module remains the audit record, while the 26-row deployable file is the asked instrument	Separates internal traceability from respondent burden.
Optimization levers	Ranking plus legacy toggle structure	Ranking plus portability bonus, redundancy control, construct coverage, documented wording replacements, and respondent-answer routing	Produces a more defensible always-on instrument.

Table 3: How the deployable questionnaire was optimized relative to the older architecture view.

7 Administration Rules

7.1 Administration sequence

1. Ask the three all-respondent Generic items.
2. Identify the active crisis context.
3. Activate one crisis-specific section based on that context and on the respondent's Generic-core answers.
4. Skip all sections that do not match the active crisis.

7.2 Respondent-answer refinement and live path lengths

Live path	Rows asked	Minutes	Calculation / rule	Why it is used and the assumption behind it
Pandemic path	8	4.32	3 Generic +5 pandemic rows and $1.40 + 2.92 = 4.32$ minutes	Used when answers or interview context indicate a pandemic/public-health event; assumes only the relevant block is activated.
Shutdown path	7	5.60	3 Generic +4 shutdown rows and $1.40 + 4.20 = 5.60$ minutes	Used for acute financial disruption or shutdown crises; assumes one dominant economic-disruption context.
Disaster path	17	21.70	3 Generic +14 disaster rows and $1.40 + 20.30 = 21.70$ minutes	Used for displacement, trauma, or direct exposure events; longest path because the disaster bank carries the deepest direct-impact content.
Full paper instrument	26	28.82	Sum of all deployable rows listed in this document	Upper bound shown for documentation; not the default live path for a single respondent.

Table 4: Respondent answers refine the effective interview length by activating one relevant crisis block.

7.3 Universe summary

Universe block	Rows	Minutes	Average R_i	Average P_i
All respondents	3	1.40	5.334	7.125
Pandemic or public-health emergency	5	2.92	3.364	4.393
Shutdown, recession, or acute financial disruption	4	4.20	3.424	4.509
Disaster, displacement, or direct exposure event	14	20.30	2.617	2.768

Table 5: Universe blocks used to route live administration.

7.4 Source contribution to the deployable instrument

Source	Rows	Minutes
Hurricane Katrina 2007	14	20.30
COVID-19	8	4.32
Govt Shutdown Income	3	2.33
Govt Shutdown Crisis	1	1.87

Table 6: Historical source contribution to the deployable questionnaire.

8 Formula Notes for Deployment

The deployable questionnaire still comes from the same scoring logic as the full codebook.

Operational formulae

$$\text{minutes} = \frac{\text{word_count} \times 7}{60}$$

$$R_i = \frac{U_i}{B_i}$$

$$P_i = \frac{\text{augmented_utility}}{B_i \times (1 + 0.65 \times \text{redundancy_scaled})}$$

- R_i explains why a question is strong relative to burden.
- P_i helps decide which similar questions survive the time-budget cut.
- Minutes is the operational field used for questionnaire assembly.

Quantity	Calculation	Why it is used	Assumption
minutes	$(\text{word_count} \times 7)/60$	Converts wording length into an operational time field for routing and budget control	Oral administration averages about 7 seconds per word.
R_i	U_i/B_i	Explains why a row is strong relative to its burden	Efficient questions should deliver more crisis signal than the time they consume.
P_i	$\text{augmented_utility}/[B_i \times (1 + 0.65 \text{ redundancy_scaled})]$	Breaks ties among similarly efficient rows by rewarding distinctiveness and portability	A short field instrument should prefer distinct rows over repeated variants.
Deployable rows	Selected scored rows minus demographic-only traceability rows	Converts the audit package into the field instrument	Demographic context is not part of the default asked path.
Live path length	Generic-core minutes plus one active crisis block	Translates the paper instrument into respondent-specific burden	Exactly one specific block is active for the standard administration path.

Table 7: How the main deployable quantities are derived and interpreted.

9 Worked Example for a Deployable Generic Item

Example question: GEN_LOSE_EARNINGS_PANDEMIC

Field	Value	Calculation	Why it is used / assumption
Question text	Did you lose earnings because of the pandemic?	Clean deployable wording retained in the universal screen	This item is portable enough to ask before any crisis-specific routing begins.
U_i	3.400	$0.80 + 0.80 + 0.90 + 0.90$	Utility aggregates matched crisis-signal weights; assumes the tagged keywords capture the substance of the item.
B_i	0.600	$\max(0.10 \times 6 + 0.20 \times 0.0, 0.1)$	Burden penalizes wording cost; assumes the fixed burden constants remain the operational design choice.
R_i	5.667	$3.40/0.60$	Shows why the row is analytically strong relative to burden.
P_i	7.860	Value computed after rarity, construct, portability, and redundancy scaling	Keeps the item ahead of lower-value or more repetitive alternatives.
minutes	0.70	$(6 \times 7)/60$	Feeds the routing and time-budget calculations; assumes 7 seconds per word in oral administration.
Administration block	All respondents	Asked before any crisis-specific skip logic	Assumes every respondent should receive a common crisis-baseline screen.

Table 8: Worked example showing how one Generic item remains short, portable, and analytically strong.

10 All-Respondent Generic Core

These questions create a portable core that can be asked before event-specific routing.

Variable	Source	Question	R_i	P_i	Minutes
GEN_EXPERIENCED_FINANCIAL_DIFFICULTIES_C	COVID-19	Have you experienced any financial difficulties because of the crisis?	5.667	7.187	0.35
GEN_LOSE_EARNINGS_PANDEMIC	COVID-19	Did you lose earnings because of the pandemic?	5.667	7.860	0.70
GEN_RECEIVE_STIMULUS_PAYMENT_OTHER	COVID-19	Did you receive a stimulus payment or other emergency government support?	4.667	6.328	0.35

Table 9: All-respondent Generic core.

11 Pandemic or Public-Health Emergency Block

Ask this section when the active crisis is a pandemic or other public-health emergency.

Variable	Source	Question	R_i	P_i	Minutes
SPC_WORK_ENTIRELY_HOME_DURING	COVID-19	Did you work entirely from home during the crisis?	3.500	3.957	0.47
SPC_WORKING_JOB_CONSIDERED_ESSENTIAL	COVID-19	Were you working in a job that was considered essential during the crisis?	4.000	5.223	1.05
SPC_RECEIVE_PAYCHECK_PROTECTION_OTHER	COVID-19	Did you receive paycheck protection or other emergency business support?	3.500	5.212	0.23
SPC_RECEIVE_STIMULUS_PAYMENT_OTHER	COVID-19	Did you receive a stimulus payment or other emergency government support?	3.500	4.466	0.23
SPC_LAID_OFF_FURLOUGHED_PANDEMIC	COVID-19	Were you laid off or furloughed because of the pandemic?	2.318	3.106	0.93

Table 10: Pandemic or public-health emergency block.

12 Shutdown, Recession, or Acute Financial Disruption Block

Ask this section when the active crisis is a shutdown, recession, or acute financial disruption.

Variable	Source	Question	R_i	P_i	Minutes
SPC_STOPPED_WORK_CRISIS	Govt Shutdown Income	Have you stopped this work because of the crisis?	3.500	4.134	0.47
SPC_WAGES_SALARY_PAYMENTS_JOB	Govt Shutdown Income	Were there any wages or salary payments from this job during the crisis period?	2.727	4.134	1.17
SPC_HOUSEHOLD_MANAGE_FINANCIAL_DIFFICULT	Govt Shutdown Crisis	How did your household manage financial difficulties caused by the shutdown or crisis?	3.719	4.592	1.87
SPC_STOPPED_WORKING_BUSINESS_CRISIS	Govt Shutdown Income	Have you stopped working at this business because of the crisis?	3.750	5.177	0.70

Table 11: Shutdown, recession, or acute financial disruption block.

13 Disaster, Displacement, or Direct Exposure Block

Ask this section when the active crisis is a disaster, displacement, or direct exposure event.

Variable	Source	Question	R_i	P_i	Minutes
SPC_HOME_DAMAGED_DESTROYED_DISASTER	Hurricane Katrina 2007	Was your home damaged or destroyed by the disaster?	2.650	2.266	1.17
SPC_BUSINESS_DAMAGED_DESTROYED_DISASTER	Hurricane Katrina 2007	Was your business damaged or destroyed by the disaster?	3.300	2.962	1.17
SPC_ANYONE_IMMEDIATE_FAMILY_KILLED	Hurricane Katrina 2007	Was anyone in your immediate family killed as a result of the disaster?	2.115	2.060	1.52
SPC_EXPERIENCE_HURRICANE_FORCE_WINDS	Hurricane Katrina 2007	Did you experience hurricane force winds at your location during the disaster?	2.077	2.106	1.52
SPC_EXPERIENCE_MAJOR_FLOODING_HOME	Hurricane Katrina 2007	Did you experience major flooding in your home during the disaster?	3.792	4.222	1.40
SPC_PHYSICALLY_INJURED_WAY_AS	Hurricane Katrina 2007	Were you physically injured in any way as a result of the disaster?	2.846	2.804	1.52
SPC_SEVERE_DAMAGE_HOME_DURING	Hurricane Katrina 2007	How severe was the damage to your home during the disaster?	2.769	3.169	1.52

Table 12: Disaster, displacement, or direct exposure block: housing, damage, and direct physical exposure items.

Variable	Source	Question	R_i	P_i	Minutes
SPC_AFRAID_DURING_DISASTER_MIGHT	Hurricane Katrina 2007	How afraid were you during the disaster that you might be seriously injured or killed?	2.447	2.584	1.87
SPC_DISASTER_OFTEN_DISTURBING_MEMORIES	Hurricane Katrina 2007	Since the disaster, how often have you had disturbing memories or images about what happened?	1.825	1.588	2.10
SPC_LONG_STAY_TEMPORARY_HOUSING	Hurricane Katrina 2007	How long did you stay in temporary housing after the disaster?	2.455	3.090	1.28
SPC_EVACUATE_HOME_DISASTER_HIT	Hurricane Katrina 2007	Did you evacuate from your home before the disaster hit?	3.227	3.530	1.28
SPC_DISASTER_CAUSE_LOSE_JOB	Hurricane Katrina 2007	Did the disaster cause you to lose your job?	3.200	3.708	1.17
SPC_DISPLACED_PLACE_LIVING_DISASTER	Hurricane Katrina 2007	Were you displaced from the place you were living because of the disaster?	2.115	2.643	1.52
SPC_RECEIVE_HELP_FEDERAL	Hurricane Katrina 2007	Did you receive any help from FEMA Federal Emergency Management Agency?	1.818	2.027	1.28

Table 13: Disaster, displacement, or direct exposure block: trauma, displacement, employment, and aid items.

14 Administration Notes

- The deployable questionnaire intentionally keeps the three **Generic** questions first because they travel best across crisis contexts.

- The deployable file excludes demographic-only rows even though those rows still appear in the master review sheet.
- Respondent answers refine the live path length: most interviews ask the 3-item Generic core plus one specific block rather than all 26 deployable rows.
- Some closely related support items appear in both the Generic and pandemic blocks because one version is meant for universal framing and the other is preserved as an event-specific probe.
- The disaster block is the longest because the Katrina source bank contributes the richest direct-exposure content.

15 Recommended Use

Use this document when you need:

1. A presentation-ready PDF version of the deployable instrument.
2. Skip logic by crisis universe.
3. A row-level record of why each question survived the selection process.

The client package is intentionally limited to the three final PDF reports, while the full traceability workbook remains an internal review artifact.